

program3.py

```
1  from collections import deque
2
3  processes = [
4      {"pid": "P1", "arrival": 0, "burst": 7, "priority": 2},
5      {"pid": "P2", "arrival": 2, "burst": 4, "priority": 1},
6      {"pid": "P3", "arrival": 4, "burst": 1, "priority": 3},
7      {"pid": "P4", "arrival": 5, "burst": 4, "priority": 2},
8  ]
9
10
11 def priority_scheduling(processes):
12     current_time = 0
13     completed = set()
14     result = []
15     while len(completed) < len(processes):
16         ready = [
17             p for p in processes
18             if p["arrival"] <= current_time and p["pid"] not in completed
19         ]
20         if not ready:
21             next_time = min(
22                 p["arrival"] for p in processes
23                 if p["pid"] not in completed
24             )
25             result.append(("IDLE", current_time, next_time))
26             current_time = next_time
27             continue
28         p = min(
29             ready,
```

python3 program3.py

INPUT DATA

PID AT BT Priority

P1 0 7 2

P2 2 4 1

P3 4 1 3

P4 5 4 2

Priority Rule: Smaller number = Higher priority

NON-PREEMPTIVE PRIORITY

Process Start End

P1 0 7

P2 7 11

P4 11 15

P3 15 16

Sequence: P1 -> P2 -> P4 -> P3

Round Robin Quantum = 2

ROUND ROBIN

Process Start End

P1 0 2

P2 2 4

P1 4 6

P3 6 7

P2 7 9

P4 9 11

P1 11 13

P4 13 15

P1 15 16

Sequence: P1 -> P2 -> P1 -> P3 -> P2 -> P4 -> P1 -> P4 -> P1

yashasvi_bhagour@LAPTOP-CJHR9PD4:~/OS-Third-Program\$